

For our Syllabus, we need to know the three hazard types, but only explore the resolution of Structural and Data Hazards.

1. **Structural Hazard:** Occurs when the system is trying to use the same resource multiple times in the **same clock cycle**.

Example-1:

C1	C2	C3	C4	C5	C6	C7	C8
IF	ID	EX	MEM	WB			
	IF	ID	EX	MEM	WB		
	IF	ID	EX	MEM	WB		

- The red rows would cause structural hazard due to same resource being used twice in the same clock cycle.

Example-2:

- Data memory and instruction memory are split so that the **IF** and **MEM** stages can exist together in the same cycle without causing a hazard.

IF stage for instruction memory and MEM stage for Data memory.

C1	C2	C3	C4	C5	C6	C7	C8	C9
IF	ID	EX	MEM	WB				
	IF	ID	EX	MEM	WB			
		IF	ID	EX	MEM	WB		
			IF	ID	EX	MEM	WB	
				IF	ID	EX	MEM	WB

2. **Data Hazard:** Occurs when an instruction depends on the completion of data access by a previous instruction.

Example: add **x19**, x0, x1
 Sub x2, **x19**, x3

	C1	C2	C3	C4	C5	C6
add	IF	ID	EX	MEM	WB	

sub		IF	ID	EX	MEM	WB
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Clock cycle 3 would cause a data hazard. Because in the **sub instruction**, the x19 register's data is decoded, but the data from the previous **add instruction** is not yet updated in the register file. It is updated in the **WB stage in clock cycle 5**.

3. **Control Hazard:** Occurs when the pipeline does not know what will be the next instruction address. It depends on the previous instruction's completion.

Example: Branch type instructions. In these instruction types, the next instruction address can have two options.

PC + Offset if the branch condition is true.

PC + 4 if the branch condition is false.

Until the instruction is fully executed (EX/ ALU), the pipeline will not know what control signals to give. This is known as Control Hazard.